DeCad | Final closure

24 October 2026 | Recovery baseline 581c236 | Model unchanged

Measurement	581c236 baseline	Closure
Clear pin-to-net score	228/245	228/245
300-DPI scan	44/72	44/72
Previously valid native pins retained	131	131
Native correct / wrong pairs	10 / 0	10 / 0
Unseen components / automatic pins	8/8; 18/19	8/8; 19/19
Raw Faculty XML exports	11/18	18/18
Backend / frontend passing tests	412 / 18	421 / 19

The automatic held-out filter now recovers all nineteen physical pin IDs. All eighteen raw Faculty XML cases export, with unresolved terminals explicitly marked for review.

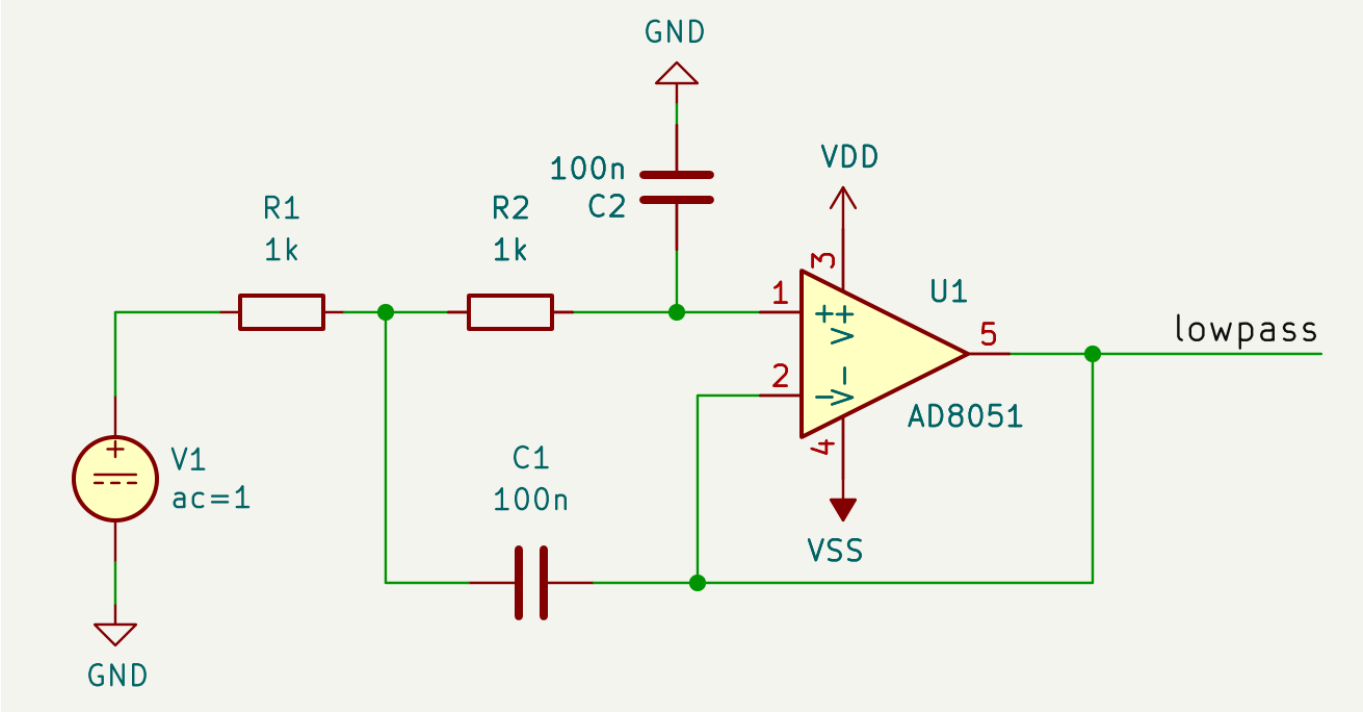
All previously correct clear assignments, native pin IDs and connected pairs remain. Provisional terminals are isolated; successful serialization is separate from electrical correctness.

The nineteenth op-amp terminal

Five visible conductors must enter the amplifier boundary: two inputs, one output and two supply leads. Four observed numbers must identify one standard five-terminal inventory. The remaining number is assigned only to the remaining visible lead. Ambiguous families and missing conductors abstain.

Unseen measurement	Literal IDs	R/C/L equivalence
Components / physical pins	8/8; 19/19	Same inventory
Correct connected pairs	6	18
Incorrect connected pairs	12	0
Missing pairs	12	0

Unprinted symmetric passive IDs follow the existing deterministic convention. Equivalence scoring permits R/C/L terminal swaps and never changes canonical export IDs. Diodes and sources are excluded.



Filter detail. The tested raster also includes the two supply-source symbols above this section.

Raw XML: review-required terminals

All 18/18 cases now serialize and roundtrip. The graph contains 37 provisional terminals, of which 0 retain explicit STUB identifiers. Each is on an isolated net and highlighted in amber in the review interface. Numbering, body ownership and net assignment require confirmation.

Case	Provisional	STUB IDs
S4-scan200	2	0
S4-scan300	6	0
S5	2	0
native-3b2a49a524e2d50dcf26	2	0
native-c4d7791b4ebdf13c3944	9	0
native-c4d88e79bc63688e0e7c	4	0
native-c6bcc506a3200b19e089	12	0

The ESP32 U4 caption occurred in prose; its complete candidate record and source region remain in quarantine. This does not establish recovery of the actual U4 device elsewhere in the design.

Native assigned physical-ID coverage is 158, including provisional numeric IDs that match the independent inventory. All 131 earlier correct IDs are retained. No provisional conductor is joined to an existing electrical net.

Verification and practical limits

Check	Result
Backend suite	421 passed, 6 skipped, 2 warnings in 445.34s (0:07:25)
Frontend suite	19 passed
All-corpus incremental checks	18 cases; cached raster postprocessing
Additional fresh OCR	6 affected cases; plus unseen and live import
Faculty XML roundtrip	18/18
KiCad / Capture native reopen	18/18 each
Literal assignments preserved	2210/2210 per CAD format
Live application	FinalClosure20261024; 19 automatic pins

Fresh affected-case OCR and cached incremental verification are reported separately. This closure did not repeat the full eighteen-case OCR run. Per-case provenance and execution providers are retained with the predictions.

A separate live provisional proposal also passed typed evidence persistence, Faculty XML export and both native download/readback paths: ClosureProvisional20261024.

Seventeen clear-source mismatches and sparse native graph coverage remain. Provisional exportability is not approval of an electrical design. Unread package numbers and ambiguous symbol ownership remain explicit review items.

Receipts: evidence/closure-20261024/verification.json; accepted/gate.json; xml-corpus/report.json; native-exports/report.json; unseen-final/report.json; live-automatic/report.json.

Protected model SHA-256: c5ad0e9ebe43373f1986ed5d6c39be0b73a29cb4dcb257e45ca7f91f41ce07cf